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A Model of the Effects of Nonuniform Soil-Water Distribution on the Subsurface Migration of Bacteria: Implications for Land Disposal of Sewage

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Abstract—Australia produces one of the highest volumes of agricultural, industrial, and municipal wastes per capita in the world. Increasingly, the public is demanding statutory authorities investigate environmentally sound methods of land application. However, a real danger associated with land disposal of wastes, in particular sewage, is the possible contamination of groundwater and threats to public health resulting from transport of pathogenic micro-organisms through the vadose zone. The study of the transport and fate of micro-organisms in soils is also of vital importance in the fields of oil recovery, biological control of plant root diseases, and in-situ bioremediation of contaminated soils and aquifers from industrial accidents. The objectives of this paper are

(a) to explore the extent of spatial and temporal heterogeneity found in the soil-water patterns in Australian soils, and
(b) to establish the framework for a mathematical model of the population dynamics and mobility of soil bacterial transport through the unsaturated zone of soils.

The model explicitly incorporates spatial and temporal heterogeneity in describing these dynamics.

Keywords—Pollution, Contamination, Sewage, Bacteria, Model.

1. INTRODUCTION

In many parts of the world, land application of waste is an economic imperative. The world population is currently 5.8 billion and by the year 2050, it is estimated to increase to 10 billion. The use of manufactured inorganic fertilisers alone will not ensure adequate sustainable agricultural production levels. The recycling of sewage and other wastes will be essential for economic and ecological survival [1]. Although the nutrient content of wastes makes them attractive as fertilisers, land application of domestic sewage, even after treatment, contains significant

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concentrations of potentially toxic contaminants including heavy metals, pesticide residues, other organic compounds such as polychlorinated biphenyls (PCBs), polychlorinated dibenzo-p-dioxins, and dibenxo-furans (PCDD/PCDFs), polycyclic aromatic hydrocarbons (PAHs), linear alkylbenzenes (LABs), fatty acids and sterols, and pathogenic organisms.

Leaching of micro-organisms through the soil is a major hurdle to the safe disposal of treated sewage on agricultural and pastured lands. For example, 51 to 55% of ten month old cattle reared on sewage-irrigated pastures near Melbourne, Victoria were found to be infected with cysticerci of Taenia saginata [2]. The study of the transport of micro-organisms is also important to other fields including the mining industry (e.g., oil recovery), biotechnology (e.g., biological control of plant root diseases), and environmental protection (e.g., in-situ bioremediation of contaminated soils or aquifers resulting from industrial accidents using indigenous or exogenous bacteria). The application of beneficial bacteria for bioremediation of contaminated sites or the biological control of plant diseases may require the release of a large number of genetically modified strains in the environment. Public concern of the environmental safety of such practices has been noted [3].

To date, only a few studies have been conducted to investigate the transport and survival of bacteria in field soils [3–7]. Fewer studies have attempted to model bacterial transport and population dynamics [8]. Modelling and monitoring transport of micro-organisms in natural and agricultural soils is difficult due to the complicated networks of interconnected pathways in the soil which can transmit water, contaminants, substrate (nutrients), and micro-organisms at varying velocities [9–12]. Preferential pathways resulting from biological and geological activity, such as subsurface erosion, faults and fractures, shrink-swell cracks, animal burrows, worm holes, decaying roots, etc., may transmit water and micro-organisms at very much higher rates than those anticipated by current theory. While the bulk of the soil matrix may act as a biological filter and impede transport of micro-organisms, preferential flow paths in soil may actively promote transport of substrate and micro-organisms which may lead inadvertently to contamination of water bodies including groundwater. The preferential flow mechanism in heterogeneous soils is well known. Recent studies with nutrients [9,10,13–17] and pesticides [18–23] have shown that solutes travel through preferential flow paths at much faster rates than anticipated by current theory. Nutrients may act as a biological substrate for micro-organisms and preferential flow paths provide greater levels of oxygen; thus the potential for transport and growth of microbial populations is high.

2. EXPERIMENTAL FRAMEWORK

2.1. Monitoring Microbial Transport in Soils

To date, most experimental programs designed to study the migration of subsurface microbial populations have been based on small, single column leaching tests. In contrast, our experimental program uses large undisturbed soil cores extracted from field sites. The advantage of taking undistributed soil columns is that they preserve the natural heterogeneity of field soils, thus offering a better means of studying preferential transport under field conditions [24]. The soil core is carefully encased in a wooden box. The interior of the box is lined with nonreactive polyurtherane foam filler to prevent deterioration of the soil structure during transport and analysis (see Figure 1). Twenty-five individual fibreglass wick lysimeters placed in a 5×5 grid cover the basal area of the sampling unit. The undisturbed soil core has a dimension of $34\,\text{cm} \times 34\,\text{cm} \times 40\,\text{cm}$ and is larger than the required basal sampling area in order to prevent possible edge effects. The wick-lysimeters are used to collect effluent and microbial populations leached from the soil core. The length and thickness of the wicks are chosen to provide a capillarity (or transport force) equivalent to that found in the soil, thus not perturbing the "natural" flow patterns found in the soil. The wick lysimeter consists of an individual wick glued to a stainless steel base plate that is firmly pressed against the soil surface by a spring. The wicks are encased with plastic tubing to prevent evaporation and bacterial contamination. Each lysimeter collects

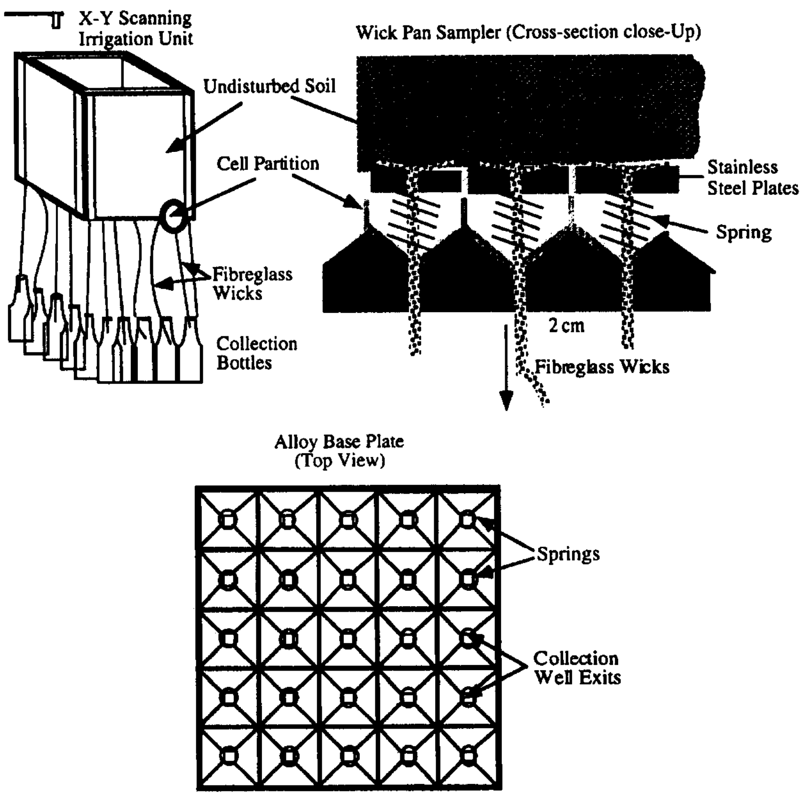

Figure 1. Multisegment percolation unit for sampling transport of soil-water, contaminants, micro-organisms, and substrate.

soil-water, chemicals, and mobile microbial populations in the neighbourhood of the wick. The soil core is irrigated by a sprinkler/drip-irrigation system and the flux controlled by a peristaltic pump. Various soil-physical properties such as hydraulic conductivity and soil-water can also be monitored with this apparatus. Soil-water probes and tensiometers can be used to detect soil-water content and suction.

2.2. Culture

Experimental protocols for the preparation, culture, and determination of "safe" bacterial populations are well documented [25]. For example, [3] used pseudomonas fluorescens marked with a spontaneous resistance to rifampin (CHA0-Rif). Reference [4] used a rifampicin resistant mutant of Burkholderia cepacia strain P2. Typically the bacteria are cultivated on King's medium B agar and left overnight at high temperature (e.g., 27°C). They are then harvested, placed in a suspension of distilled water and sieved through several layers of cheesecloth to select a uniform cell density. The bacterial population of known concentration are then drip irrigated on the soil surface. A simple procedure for the enumeration of bacteria in the effluent using an ion-selective electrode is described in [3].

3. EVIDENCE OF PREFERENTIAL TRANSPORT IN AUSTRALIAN SOILS

The first step in the investigation of possible locations for the safe disposal of sewage and wastes is site evaluation for the propensity of preferential transport and risk of groundwater contamination. Several Australian soil types have been chosen for investigation and are briefly described below. The field sites are named as:

(a) Redland Bay,

(b) Tower Hill,

(c) Grassmere, and

(d) Rutherglen.

The first site (Redland Bay) is located on the coastal fringe of Moreton Bay, approximately 30 km south-east of Brisbane, Queensland. The Redland Bay region consists of extensive market-gardens and is a major source supplying vegetables and fruit to local markets in Brisbane. The region experiences frequent tropical storms in summer with heavy intensity, high volume, rainfall. In contrast, the winter months experience in-frequent, low-volume, and low-intensity rainfall. The soil of this region is described as a Redland Bay Krasnozem, a clay and organic rich, red-earth soil. The extracted soil column had noticeably large pebbles ranging from 1 to 5 mm randomly distributed throughout the column. The other sites are located in different regions in the state of Victoria. The Tower Hill site is located in a state game reserve in the Western District of Victoria. This site is located within a dormant volcanic crater. The reserve is surrounded by market gardens growing predominantly potatoes, carrots, and onions. An underlying upper tertiary limestone aquifer, used for both domestic and stock purposes, is known to have a large number of fractures which makes preferential flow highly likely. The soil column was extracted from a ridge on the crater's tuff. The soil is described as a grey, clay loam with a nearly uniform distribution of particle sizes and moderate organic content (about 3–5% by dry-weight).

Two large undisturbed soil cores were taken from a farm located at Grassmere approximately 40 km north of Warrnambool, 300 km west of Melbourne in the Western District of Victoria. The experiments are referred to as Grassmere-1 and Grassmere-2. The Grassmere site lies in a highly productive beef and dairy farming area. The area is situated on gently undulating basalt plains. Bedrock consists of olivine and iddingsite basalt, limburgite, minor scoria, and tuff. Soil above the bedrock consists of fine soil particles through to clay with buckshot inclusions (up to 1 cm diameter) and randomly distributed rocks varying in diameter from a few millimetres to approximately 20 cm. The soil is described as a dark balsaltic soil with high organic content (approximately 10%). Two large undisturbed soil cores were also extracted from the Rutherglen site. Rutherglen is located about 350 km north of Melbourne. The region is noted for sheep and beef cattle grazing. The site is located in the Rutherglen Experimental Farm operated by Agriculture Victoria. The experimental farm is trying a large-scale, intensive beef feedlot for the purpose of developing management guidelines for the application of manure and urine on surrounding agricultural land. The soil is a red clay-loam. The two experiments conducted on the extracted soils from this site are referred to as Rutherglen-1 and Rutherglen-2, respectively. All the sites were selected on the basis of their importance to local economies and potential threat to the environment by surface application of chemicals and wastes.

High volume rainfall, typical of tropical summer rainfall was applied to the Redland Bay soil column. The soil was continuously irrigated at a rate of approximately 13 mL/min for 25 days commencing on November 14, 1994. The soil surface was irrigated using a rain simulator consisting of 81 hypodermic syringes arranged on a regular 9×9 grid pattern. The syringes were located on the base of a small reservoir ($25 \times 25 \times 1$ cm) which was connected to a peristaltic pump. The reservoir was suspended above the soil column and moved laterally with the aid of two small electric motors, thus ensuring uniform irrigation on the soil surface. A similar rainfall applicator was used to irrigate the Tower Hill soil. The irrigation experiment was conducted over a three week period in September 1995. The application rate was approximately 12–14 mL/min, which simulates a high winter rainfall rate for this region. In 1996, an improved rain simulator was developed using an x-y scanning drip irrigation system. This irrigator was used to apply rainfall to the Grassmere 1 and 2 and the Rutherglen 1 and 2 cores. For these cores, the application rate was approximately 3 mL/min. The Grassmere-1 core was irrigated for 66 days commencing in

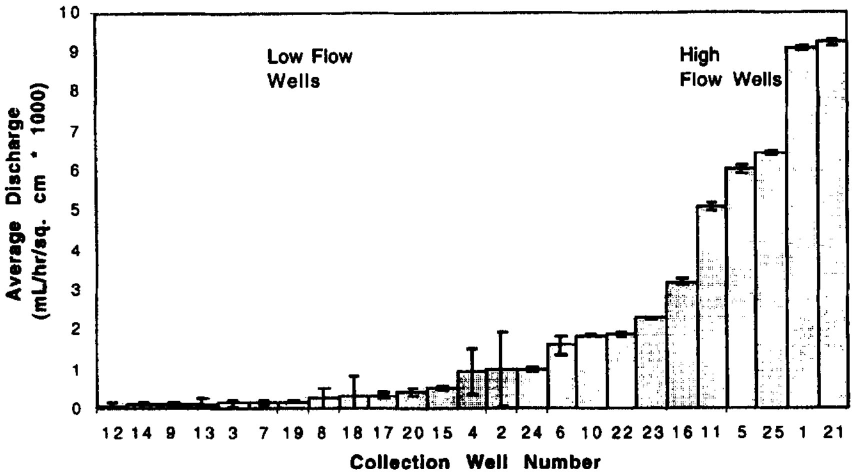

Figure 2. Preferential percolation of soil-water from the Rutherglen-2 soil. Data were collected over a 66 day period. The error bars are intervals of plus and minus one standard error.

August 1996. The Grassmere-2 core was irrigated for 18 days commencing in October 1996. The Rutherglen-1 core was irrigated for seven days commencing in October 1996 and the Rutherglen-2 core was irrigated for 27 days in September 1996.

Figure 2 illustrates typical small-scale spatial variation in drainage patterns for the Rutherglen-2 soil core. The collection well number refers to the individual lysimeter collecting the eluted volume from the base of the core. In spite of a uniform application of irrigation on the soil surface, the drainage eluted from the base of the core is very highly variable (one-way ANOVA, $p < 0.0001$). For this case, as much as a 14-fold variation in the drained volume was observed. Similar patterns in drainage for the other experimental sites were also observed, indicating significant heterogeneity in chemical and physical properties of the soils and potential risk of rapid transport of micro-organisms.

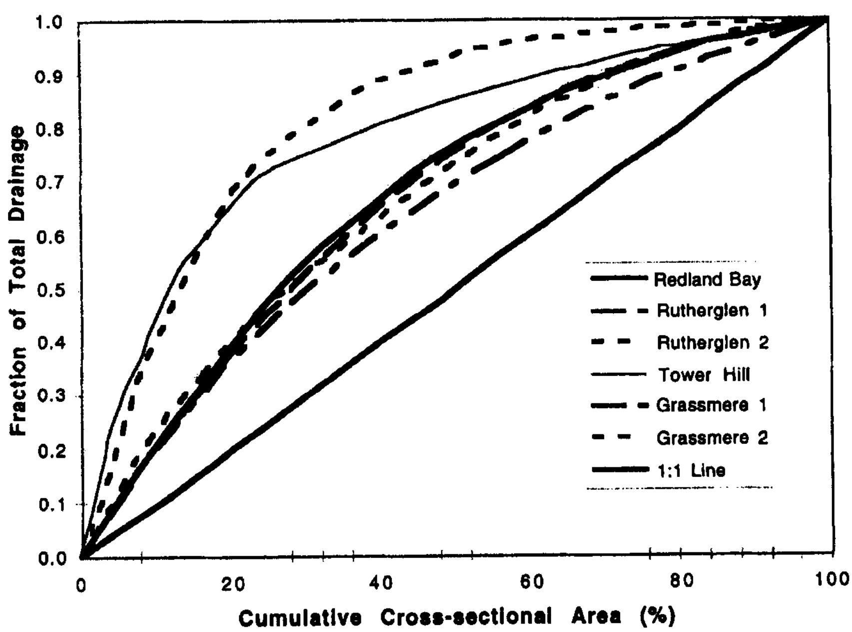

Figure 3. Spatial variation in drainage patterns for all experiments. The fraction of the total drainage is plotted as a function of the cumulative cross-section of the area of the collection base.

Figure 3 illustrates the spatial variation in drainage patterns for all soil experiments. The fraction of the total drainage for each collection well (lysimeter) was calculated and these values were then sorted and ranked in descending order and plotted with the cumulative cross-sectional area of the soil core. If the soil core were draining soil-water at a uniform rate, then there would be no spatial variation in the drainage volume collected by each well. Each collection well would contribute an equal volume of soil-water to the total drainage. In other words, if drainage were uniform, the fraction of the total drainage plotted with the cumulative cross-sectional area would fall on a 1 to 1 line. Departures from a 1 to 1 line indicate heterogeneity or preferential flow. As indicated in the figure, the Rutherglen-2 and Tower Hill experiments exhibit very strong preferential flow, e.g., about 70% of the drained soil-water was collected from less than 25% of the area of the base of the core. The other experiments exhibit moderate preferential flow. For these soil cores, approximately 70% of the eluted water was drained by 40% of the basal area. The implications of these results are obvious; nutrient, substrate and bacteria mobilised in the soil-water may be transported preferentially through the soil.

Other studies of nutrients and contaminants have shown that the preferential transport may occur at much greater soil depths and at far more rapid times than current models predict [5,14,26–44].

4. A MATHEMATICAL MODEL FOR PREFERENTIAL FLOW OF MICRO-ORGANISMS

The experimental program is designed to gain a clear understanding of the physical and biological processes of transport of micro-organisms in natural and agricultural soils under conditions of preferential flow. Current theories of the mobility and population dynamics of soil bacteria and other microbial populations neglect preferential flow and consequently are inappropriate for application to field sites. Therefore, a mathematical model that explicitly incorporates the spatial and temporal heterogeneity of soil-water distribution is required for predicting the concentration of soil microbial populations in field soils [10].

The transport of micro-organisms in and out of the liquid phase is controlled by the soil-water flux that results in advective transport, hydrodynamic dispersion and diffusion; and in the case of motile micro-organisms by their own active movement. In structured porous media with cracks and channels, advective transport may be also highly variable. Micro-organisms are also subjected to biological and physio-chemical forces that influence their movement. These forces produce mechanisms of retention that include filtering (or straining), adsorption, interception, attachment, and sedimentation. Micro-organisms are also subject to growth and decay and are affected by growth-limiting substrates that are simultaneously transported in the soil. All these factors have to be considered when constructing a comprehensive mathematical description of microbial transport [8]. The mathematical formulation for the advection, mixing, diffusion, growth, and decay of micro-organisms in porous media were previously described by [8]. We shall adopt their basic formulation, and where appropriate, modify it for preferential transport.

4.1. Advection

Moving soil-water carries with it dissolved gases, solutes, and suspended colloidal material such as particles, organic matter, bacteria, and viruses. The mass flux of micro-organisms carried in soil water is proportional to the concentration of micro-organisms in solution. This may be described by the following equation [8]:

$$J_v = J_w C = v_w \theta C, \tag{1}$$

where J_v is the advective flux of micro-organisms, J_w is the water flux, C is the concentration of micro-organisms, v_w is the average water velocity, and θ is the soil-water content.

4.2. Chemotaxis

The systematic active movement of micro-organisms to chemical gradients is called chemotaxis. Chemotactic micro-organisms move toward chemical attractants and away from toxins. The average chemotactic velocity, v_x, is defined by

$$v_x = \left(\frac{k_x}{S}\right) \nabla S, \tag{2}$$

where k_x is the chemotactic coefficient and S is the concentration of the substrate. The macroscopic flux due to chemotaxis is, therefore, given by

$$J_x = v_x \theta C. \tag{3}$$

Note, for viruses, $J_x = 0$. Usually for other micro-organisms, the chemotactic flux is small in comparison with other transport fluxes and can be ignored [8].

4.3. Diffusion, Hydrodynamic Dispersion, and Motility

Nonuniform concentration of micro-organisms are expected in natural soils. Diffusion acts to equalise the spatial distributions of micro-organisms. The diffusion flux can be described by Fick's first law. Diffusion occurs whether the soil-water is flowing or stagnant within the soil matrix. In contrast, hydrodynamic dispersion that is caused by advection not only affects the vertical migration of micro-organisms but also mixing. A third flux results from motile micro-organisms (i.e., organisms propelled by their own flagella, etc). Motile micro-organisms tend to exhibit chaotic movement and can be modelled by Brownian diffusion. Although these three processes are different, in practice it is difficult to distinguish them experimentally, and therefore, for the purposes of modelling, these three processes may be considered together as described by the following equation [8]:

$$J_d = -\tau \theta D \nabla C, \tag{4}$$

where J_d is the combined diffusion-dispersion flux, D is the diffusion-dispersion coefficient, and τ is the pore-space tortuosity.

4.4. Growth and Decay

Although bacteria can utilise several substrates, reproduction rate is often controlled by a single rate-limiting substrate and growth is controlled by the concentration of the substrate. A modified form of the Monod equations [8] is adopted here,

$$R_g^w = \frac{\mu_m S \theta C}{[K_s + S]} \quad \text{and} \quad R_g^a = \frac{M L_b \rho_b \mu_m S \eta}{[K_s + S]}, \tag{5}$$

where R_g is the microbial growth rate in the water phase (superscript w) or in the attached biomass phase (superscript a), μ_m is the maximum specific growth rate, S is the concentration of the rate-limiting substrate, K_s is the saturation constant (concentration of substrate at which specific growth rate is half the maximum value), M is the surface area of the biofilm-water phase boundary per unit volume of soil, L_b is the biofilm thickness, ρ_b is the bacteria density, and η is an effectiveness factor (which corrects for diffusional resistance at the biofilm-water phase boundary). The decay of bacteria (or inactivation of viruses) is usually described by an irreversible first-order rate equation, expressed as

$$R_d^w = \beta \theta C \quad \text{and} \quad R_d^a = \beta \rho_s C^a, \tag{6}$$

where R_d represents decay or deactivation in the water phase (w) or the attached phase (a), β is a decay-deactivation coefficient, and C^a is concentration of micro-organisms in attached biomass phase. Predation can also affect decay and a predator-prey relationship can easily be incorporated into the above descriptions.

4.5. Adsorption, Filtering, Sedimentation, and Interception

Adsorption is a complex process that refers to the "attraction" of micro-organisms to solid particles in the soil matrix or in colloidal suspension. Adsorption of micro-organisms is dependent on many factors including species type and whether the soils are hydrophobic (water-repellent) [41,45]. Filtering, as the name suggests, refers to straining particles from a suspension. The rate of filtering in heterogeneous soils with preferential flow paths could be a significant factor. Sedimentation is the settling out of particles in suspension due to gravity. In principle, sedimentation of micro-organisms is possible but is considered inconsequential, particularly in heterogeneous porous media. Interception refers to the deposition of micro-organisms resulting from collisions between the soil particles. It affects retention (retardation) but alone is unlikely to be a significant factor in transport. All these processes contribute to attachment and detachment. Attachment and detachment processes are either kinetic (nonequilibrium) or equilibrium [8]. Like solute transport, equilibrium processes of attachment and detachment of micro-organisms (caused by adsorption) can be described by linear or Freundlich isotherms [46]. Using Freundlich isotherms, the difference between the rate of attachment R_a and the rate of detachment R_b is given by

$$R_a - R_b = \rho K N C^{N-1} \frac{\partial C}{\partial t}, \tag{7}$$

where ρ is the soil bulk density, and k and N are empirical constants which depend on species and soil type. Nonequilibrium processes include filtering, interception, sedimentation, and kinetic adsorption. All these processes are considered together and mathematical expressions are easily developed. These expressions are usually empirical. Tan and Bond [8] proposed the following model for attachment:

$$R_a = K_a \theta C \left[\frac{1 - C^a}{C_m^a} \right] \tag{8}$$

and for detachment [8] recommend the model of Taylor and Jaffe,

$$R_a = \beta \rho_b C^a + \frac{\beta' M L_b \rho_b \mu_m S \eta}{K_s + S}. \tag{9}$$

The transport of micro-organisms in soils is based on a continuity equation which can be written as

$$\frac{\partial(\theta C)}{\partial t} + \frac{\partial(\rho_s C^a)}{\partial t} = -\nabla \bullet J + R_g - R_d, \tag{10}$$

where the combined flux of micro-organisms is given by

$$J = v\theta C - \tau \theta D \nabla C \tag{11}$$

and where v is the combined water flux. The rate of change of immobile or attached micro-organisms can be described by

$$\frac{\partial (\rho_s C^a)}{\partial t} = R_a - R_b + R_g^a - R_d^a. \tag{12}$$

4.6. Preferential Soil-Water Transport

The water flux v described in equation (11) can be modelled by modifying the multidomain preferential transport model developed by [12,17]. Soil-water may travel through a soil by any number of possible routes and at differing velocities. Thus the residence times of water, substrate, and micro-organisms in the vadose zone is highly variable. However, for the purposes of modelling, even though these paths may be highly distributed throughout the soil, they may be considered as groups of pathways or 'capillary-bundles' in which the soil-water and micro-organisms are transported with approximately the same flux, J as described in equation (11). These 'capillary-bundles' are also termed 'pore-groups'. The total amount of soil-water, $\theta(x,t)$ in the soil at time, t, and location, x, is the sum of all individual soil-water contents for each capillary-bundle, p (where θ_p is the individual soil-water content for the p^{th} pore-group).

$$\theta(x,t) = \sum_{p=0}^{N} \theta_p(x,t). \tag{13}$$

The maximum amount of soil-water that each group can hold and transmit is defined by $\Delta M_p = M_p - M_{p-1}$, where M_p and M_{p-1} are various soil-water contents representing upper and lower limiting values for the p^{th} group. The maximum amounts of soil-water, ΔM_p, are functions of the size of pores in each group. When $\theta_p = \Delta M_p$, then all the pathways in the p^{th} group are completely saturated with soil-water.

The soil is locally saturated when all flow paths in all groups are saturated, i.e., $\theta(x,t) = \theta_s$ and $\theta_p(x,t) = \Delta M_p$, for all p. θ_s is the saturated soil-water content of the soil. The group's soil-water content, θ_p, is a function of the vertical percolation rate, q_p, and the effects of precipitation, evapotranspiration and loss or gain of soil-water from interactions and exchanges with other groups. Therefore, by analogy to equation (11), the transport equation for each pore-group may be written as

$$\frac{\partial[\theta_p(x,t)]}{\partial t} + \frac{\partial[q_p(x,t)]}{\partial x} = A_p(x,t) \text{ for } t \geq 0; \quad x \geq 0; \quad 0 \leq \theta_p \leq \Delta M_p; \quad p = 0, \ldots, N, \tag{14}$$

where $A_p(x,t)$ is a source/sink term representing the effects of precipitation, evapotranspiration, and mixing between other groups, t is time, and x is a vertical distance with $x = 0$ being the soil surface, N is the number of mobile water pore-groups. From consideration of Darcy's law,

$$\begin{gathered} q_p(x,t) = k(\theta_p), H(x,t) \text{ for } p = 1, \ldots, N; \qquad q_0(x,t) = 0 \text{ for } p = 0; \\ \text{and } q_0 \leq q_1, \ldots, q_N, \end{gathered} \tag{15}$$

where, $H(x,t)$ is the hydraulic gradient and $k(\theta_p)$ is the hydraulic conductivity. For soil-water in the unsaturated zone, an often satisfactory assumption is to assume a unit gradient, i.e., $H(x,t) = 1$. This is a useful and accurate approximation when gravity is the main driving force, and therefore, after substitution

$$\frac{\partial \theta_p}{\partial t} + \nu_p \frac{\partial \theta_p}{\partial x} = A_p, \qquad \text{where } \nu_p = \frac{dk(\theta_p)}{d\theta_p}, \tag{16}$$

where ν_p is the velocity of percolating soil-water in the p^{th} group. Note that for $p = 0$, $\nu_p = 0$. The solution of equation (16), obtained by the method of characteristics, is given by

$$x(t) = \nu_p t + x_0 \text{ and } \theta_p(x,t) = \int_0^t A_p \, dt + \theta_p(x_0, 0); \qquad x \geq \upsilon_p t. \tag{17}$$

Equation (17) cannot be easily integrated because A_p is generally not a simple analytical expression. Therefore, a fixed time-step numerical solution for the soil-water content θ_p is required. From equation (17), in a small time interval, Δt, soil-water will travel a distance, Δx_p, as described by

$$\theta_p(x,t) = \theta_p(x - \Delta x_p, t - \Delta t) + I_p(x,t); \qquad \Delta x_p = \nu_p \Delta t, \tag{18}$$

where I_p is the integral of A_p between $(t - \Delta t)$ and t. The quantity, Δx_1 is the distance travelled by soil-water in the slowest moving group (i.e., $p = 1$). Let Δx_p be a simple integer multiple γ_p

of Δx_1, i.e., $\Delta x_p = \gamma_p \Delta x_1$. Note that in the "smaller" time step $(\Delta t/\gamma_p)$, the p^{th} group moved a distance Δx_1. Therefore, define a grid of points in $(x - t)$ space based on Δx_1 and $(\Delta t/\gamma_T)$ where $T = N + 1$, is the label for the pore-group with the greatest flux. Then $x = i\Delta x_1$; $i = 0, 1, 2, \ldots, L/\Delta x_1$ and $t = j(\Delta t/\gamma_T)$; $j = 0, 1, 2, \ldots$, and where L is the positive depth to groundwater measured from the soil surface. On this grid, only one of two parameters, either the spatial coordinate, Δx_1 or the time coordinate $\Delta t/\gamma T$ needs to be specified. It is usually more convenient to fix the time step rather than the distance, and therefore, the following equation for soil-water transport results.

$$\theta_p^{(i,j)} = \theta_p^{(i-\delta_p\gamma_p, j-1)} + I_p^{(i,j)}; \qquad \delta_p = \begin{cases} 0, & j \bmod \gamma_p \neq 0, \\ 1, & j \bmod \gamma_p = 0. \end{cases} \tag{19}$$

The total number of pore-groups, the amount of water in each group and its flux cannot be individually measured by experiments and generally will have to be estimated or determined by calibration. Indeed, it is difficult to conceive of an experiment that could determine such groupings. However, [12] proposed a simple but general method for selecting any number of pore-groups for a soil based on a piece-wise linear approximation of the hydraulic conductivity function.

The concentration of micro-organisms, $C(x,t)$, in the soil is given by $N(x,t)/\theta(x,t)$, where

$$N(x,t) = \sum_{p=1}^{T} n_p(x,t) \quad \text{and} \quad n_p^{(i,j)} = n_p^{(i-\delta_p\gamma_p, j-1)} + \Psi_p^{(i,j)} \tag{20}$$

and where $N(x,t)$ is the total microorganism population in time t at location x, n_p is the population in pore-group p and $\Psi_p^{(i,j)}$ is the change in population in the p^{th} group during $(\Delta t/\gamma_T)$ resulting from exchange (motility and dispersion), death, birth, adsorption, etc., as described by equations (5)–(9).

5. CONCLUSION

Soils often exhibit significant soil-physical and chemical heterogeneity. Numerous experiments have illustrated that soil-water flow patterns are highly variable resulting in preferential flow of nutrients, pesticides, and contaminants. It is likely that soil-bacteria may also be transported to much greater depths and at greater concentrations that would normally be anticipated by current theory. It is, therefore, necessary to construct a mathematical model that specifically addresses soil-water heterogeneity to effectively predict subsurface microbial populations. The processes controlling the transport and distribution of micro-organisms in soil-water and the soil-matrix has been outlined and a mathematical model of preferential transport of bacteria has been developed. Future research requires careful monitoring of microbial populations in undisturbed soil cores and field soils. Monitoring and modelling microbial populations is vital to the successful implementation and operation of management practices designed for land disposal of sewage.

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